Multi-Site VLAN Network with Inter-Router Routing & ACL Security

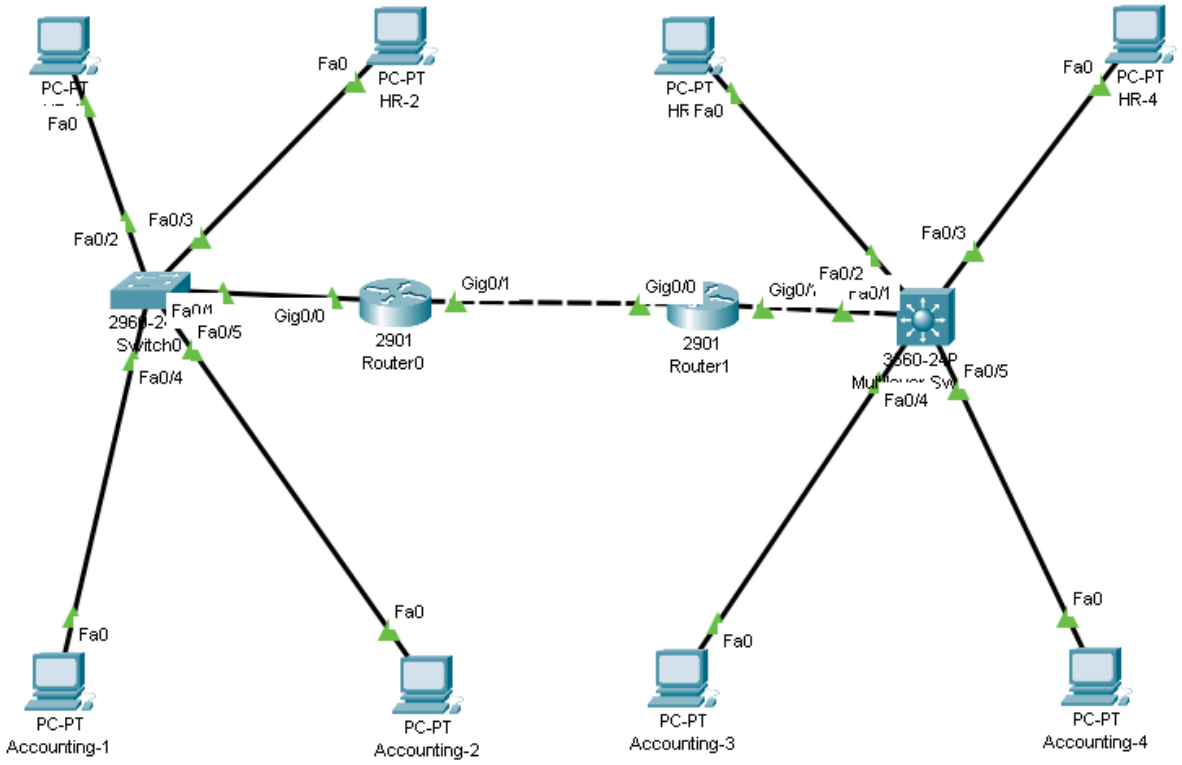
Cisco Packet Tracer · Jorge Antonio Nava · May 1st, 2026

Project Overview
Designed and configured a dual-site enterprise network in Cisco Packet Tracer featuring VLAN segmentation, router-on-a-stick inter-VLAN routing, point-to-point WAN links, and Access Control Lists to enforce traffic policies. The network spans two LANs: one using a Layer 2 2960 switch with a dedicated router, and one using a Layer 3 3560 multilayer switch. Both connected via a simulated WAN link with full cross-site reachability for same-VLAN devices.

Tools & Environment
Simulated Platform - Cisco Packet Tracer 8.x
Routers - 2x Cisco 2901 (Router0, Router1)
Switches - Cisco 2960 (Layer 2), Cisco 3560 (Layer 3 multilayer)
End devices - 8x PC-PT
Protocols & features - 802.1Q trunking, Router-on-a-Stick, Static Routing, Extended ACLs
Addressing - IPv4, /24 subnets per VLAN, /30 WAN point-to-point link

Goal:

- Segment HR (VLAN10) and Accounting (VLAN20) into isolated broadcast domains across both sites
- Allow same-VLAN devices across Router0 and Router1 to communicate freely (HR↔HR, Accounting↔Accounting)
- Block all cross-VLAN traffic — HR cannot reach Accounting and vice versa, regardless of site
- Demonstrate router-on-a-stick on a Layer 2 switch side and SVI-based routing on the Layer 3 switch side



Key configurations demonstrated

Router-on-a-Stick subinterfaces | dot1Q trunk ports | Extended ACLs | Static routing | Inter-VLAN routing | SVI configuration | Point-to-point /30 WAN | DNS & HTTP server | Default gateway design | VLAN access ports | ip routing (L3 switch) | Troubleshooting with show commands

Configurations

Router0 CLI

Enable interface GigabitEthernet0/0- Router0 to Switch0 Connection

```
interface GigabitEthernet0/0
 no shutdown
```

Create Subinterfaces for VLAN10 and VLAN20

```
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
 no shutdown

interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
 no shutdown
```

Enable WAN Interface Connection

```
interface GigabitEthernet0/1
 ip address 10.0.0.1 255.255.255.252
 no shutdown
```

Enable ISP or Next Hop IP to Router1

```
ip route 0.0.0.0 0.0.0.0 10.0.0.2
```

Switch0 (2960) CLI

Create Trunk Port Connected to Router0

```
interface FastEthernet0/1
switchport mode trunk
```

Create and Enable Access Ports for VLAN 10 and VLAN 20

```
interface FastEthernet0/2
switchport mode access
switchport access vlan 10

interface FastEthernet0/3
switchport mode access
switchport access vlan 10


interface FastEthernet0/4
```

```
switchport mode access
switchport access vlan 20

interface FastEthernet0/5
switchport mode access
switchport access vlan 20
```

*Also make sure the VLANs actually exist on the switch:

```
vlan 10
name HR-VLAN10

vlan 20
name Accounting-VLAN20
```

Router1 CLI

Assign Interface, IP Address, and Subnet Mask for Router1

```
interface GigabitEthernet0/0
 ip address 10.0.0.2 255.255.255.252
 no shutdown
```

Enable GigabitEthernet0/1 Interface (Router1 Link Connected to Switch1), Don't Assign IP Address

```
interface GigabitEthernet0/1
 no shutdown
```

Create and Assign Interfaces and Default Gateways for VLAN 10 and VLAN 20

```
interface GigabitEthernet0/1.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
 no shutdown

interface GigabitEthernet0/1.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
 no shutdown
```

Enable ISP or Next Hop IP to Router0

```
ip route 0.0.0.0 0.0.0.0 10.0.0.1

no router rip
```

***make sure Fa0/1 (uplink to Router1) is a trunk port so VLAN-tagged traffic can pass.**

Multilayer Switch (3560) CLI

Enable Routing for Multilayer Switch

```
ip routing
```

Create VLAN 10 and VLAN20 and Add SVI Gateways for both VLANs

```
vlan 10
name HR-VLAN10

vlan 20
name Accounting-VLAN20
```

Enables Trunk Port for MultiLayer Switch Connected to Router1

```
interface FastEthernet0/1
switchport trunk encapsulation dot1q
switchport mode trunk
```

Create and Assign VLAN 10 and VLAN20

```
interface FastEthernet0/2
switchport mode access
switchport access vlan 10

interface FastEthernet0/3
switchport mode access
switchport access vlan 10

interface FastEthernet0/4
switchport mode access
switchport access vlan 20

interface FastEthernet0/5
switchport mode access
switchport access vlan 20
```

End User PCs

Make sure each PC's **Default Gateway** is set correctly:

- HR PCs → `192.168.10.1`
- Accounting PCs → `192.168.20.1`

Verify After Fixing

- show ip interface brief ← all interfaces should be up/up

- show ip route ← should show a default route S* 0.0.0.0/0
- show interfaces trunk ← on the switch, confirms trunk is up

***From an end user PC, ping the default gateway first, then ping the web server. Work one hop at a time to determine whether connectivity breaks.**

Inter-VLAN Traffic Control using Access Control Lists (ACLs)

Right now, VLAN10- HR and VLAN20- Accounting are speaking freely to each other which defeats the purpose of a segmented, multi-VLAN environment. I need to block communication between these VLANs while allowing seamless communication between end devices on the same VLAN.

Goal

Allow Connections:

- VLAN 10 (192.168.10.0/24)<---> VLAN 10 on Both Router0 and Router1 sides
- VLAN 20 (192.168.20.0/24)<---> VLAN 20 on Both Router0 and Router1 sides

Block Connections:

- VLAN10 talking to VLAN20 on both sides of the Routers
- VLAN20 talking to VLAN10 on both sides of the Routers

Issues

- Hr and Accounting from Router 0 cannot speak to HR and Accounting from Router1

Solution

- Give each side unique subnets
- VLAN10 on Router0 = .10 Subnet Remains
- VLAN20 on Router0= .20 Subnet Remains
- VLAN10 on Router1 = Change to 192.168.30.X Subnet
- VLAN20 on Router1 = Change to 192.168.40.X Subnet

Location	VLAN	New Subnet	Gateway
Router0 side HR	VLAN10	192.168.10.0/24	192.168.10.1
Router0 side Accounting	VLAN20	192.168.20.0/24	192.168.20.1
Router1 side HR	VLAN10	192.168.30.0/24	192.168.30.1
Router1 side Accounting	VLAN20	192.168.40.0/24	192.168.40.1

New Goal

- HR (Router0) can communicate with HR (Router1)
- Accounting (Router0) can communicate with Accounting (Router1)

Assign New SVI Gateways on Router1's LAN.

Adding unique subnets 192.168.30.X for VLAN10 and 192.168.40.X for VLAN20 ensures that end user devices will communicate with devices on Router0 that are on the same VLAN

Router1 CLI

Assign Interfaces and New IP Address to Allow Communication between HR on Router0 and Router1

```
interface GigabitEthernet0/1.10
no ip address
ip address 192.168.30.1 255.255.255.0
```

Assign Interfaces and New IP Address to Allow Communication between Accounting on Router0 and Router1

```
interface GigabitEthernet0/1.20
no ip address
ip address 192.168.40.1 255.255.255.0
```

Add static routes so Router0 knows where Router1's subnets are:

```
ip route 192.168.30.0 255.255.255.0 10.0.0.1
ip route 192.168.40.0 255.255.255.0 10.0.0.1
```

Router0 CLI

Add static routes so Router0 knows where Router1's subnets are:

```
ip route 192.168.30.0 255.255.255.0 10.0.0.2
ip route 192.168.40.0 255.255.255.0 10.0.0.2
```

Router0 CLI

Update the ACLs on Router0

Now update both routers to block cross-VLAN traffic using the new subnets:

```
no ip access-list extended BLOCK_INTERVLAN

ip access-list extended BLOCK_INTERVLAN
deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
deny ip 192.168.10.0 0.0.0.255 192.168.40.0 0.0.0.255
deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
permit ip any any

interface GigabitEthernet0/0.10
ip access-group BLOCK_INTERVLAN in

interface GigabitEthernet0/0.20
ip access-group BLOCK_INTERVLAN in
```

Router1 CLI

Update the ACLs on Router1

Now update both routers to block cross-VLAN traffic using the new subnets:

```
no ip access-list extended BLOCK_INTERVLAN

ip access-list extended BLOCK_INTERVLAN
deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
deny ip 192.168.30.0 0.0.0.255 192.168.40.0 0.0.0.255
deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255
deny ip 192.168.40.0 0.0.0.255 192.168.30.0 0.0.0.255
permit ip any any

interface GigabitEthernet0/1.10
ip access-group BLOCK_INTERVLAN in

interface GigabitEthernet0/1.20
```

ip access-group BLOCK_INTERVLAN in

Update All PCs on Router1's Side

Every PC connected to the 3560 switch needs new IPs and gateways:

PC	New IP	Gateway
HR-3, HR-4	192.168.30.x	192.168.30.1
Accounting-3, Accounting-4	192.168.40.x	192.168.40.1

What This Achieves

Devices on the same VLAN can communicate regardless of which LAN they are on

- HR-Router0 (10.x) → HR-Router1 (30.x) allowed
- Accounting-Router0 (20.x) → Accounting-Router1 (40.x) allowed

Devices on different VLANs cannot communicate with each other regardless of which LAN they are on

- HR-Router0 (10.x) → Accounting anywhere (20.x or 40.x) blocked
- Accounting-Router0 (20.x) → HR anywhere (10.x or 30.x) blocked

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Troubleshoot Issues- The Problems Diagnosed

Router0 — 3 Problems

1. No subinterfaces — it's a flat interface, not router-on-a-stick

- `GigabitEthernet0/0` has a single IP `192.168.10.1` with no VLAN encapsulation. It can only serve one VLAN.
- VLAN20 (Accounting) has no gateway at all.

2. `GigabitEthernet0/1` has no IP address

- This port is likely an uplink toward the internet or another router, it needs an IP.

3. No default route

- There is no `ip route 0.0.0.0 0.0.0.0` anywhere. Traffic destined for the internet has nowhere to go.

Switch0 (2960) for Router0 — 2 Problems

4. Fa0/1 (the uplink to Router0) is a plain access port with no VLAN assignment

- For router-on-a-stick to work, the port connecting the switch to the router **must be a trunk**.
- Right now it's just a default access port on VLAN1, so VLAN10 and VLAN20 traffic never reaches the router.

5. The switch has SVI (Layer 3) addresses configured — but the 2960 can't route.

- You have `interface Vlan10 ip address 192.168.10.1` and `interface Vlan20 ip address 192.168.20.1` on a 2960.
- The 2960 is a Layer 2 only switch, it cannot route between VLANs. These SVIs don't do inter-VLAN routing; that job belongs to Router0.

Router1 — 2 Problems

6. `GigabitEthernet0/1` has IP `192.168.10.1` but no subinterfaces

- Same problem as Router0. One physical IP can't serve two VLANs. VLAN20 has no gateway.

7. No default route, and RIP is configured but empty `router rip` is there but has no `network` statements, so it's doing nothing.

- Need a default route.

3560 Multilayer Switch (Router1) — 1 Problem

8. `ip routing` is not enabled, and no SVI gateways are configured

- The 3560 *can* do Layer 3 routing, but you haven't enabled it or given the VLANs SVI addresses.
- The ports for VLAN10/20 have no gateway reachable by end users.

Solutions

Two Valid Switches on Each Router

- On Router0's LAN, I implemented a Layer 2 2960 Cisco switch that will handle inter-VLAN routing through its assigned subinterfaces. Router0 will also play an important role in knowing which ip subnet will be routed according to the current configurations
- On Router1's LAN, I implemented a Layer 3 3650 Cisco Switch, a multilayer switch that can route VLANs itself while Router1 only routes WAN/ internet connection, reducing VLAN routing and focusing on ensuring seamless internet connectivity.

Switches	How it works
2960 Router-on-a-Stick	Router does inter-VLAN routing via subinterfaces; switch stays Layer 2
3560 Switch routing	The 3560 routes between VLANs itself; router only handles WAN/internet

- For Router0's side (2960 switch), you **must** use router-on-a-stick since the 2960 can't route.
- For Router1's side (3560), you should use the multilayer switch to route locally.

Notes/ Enhancements to Consider

- Add Trusted zone internal network= firewalls, servers, internal devices, etc…

References

- How to set ip address in cisco switch packet tracer (layer 3 switches)? How to set ip address in Switch Cisco packet tracer? - California Learning Resource Network

- Cisco Packet Tracer VLAN Configuration Example VLAN Configuration on Cisco Packet Tracer - 2026 Updated